

Trade Rationale, Historical Analysis, and Risk Profile

We evaluate a one-month SPY risk-reversal initiated on 9 April 2025. The structure consists of selling 1000 contracts of the 95% moneyness put (strike approximately 480) and buying 1000 contracts of the 105% moneyness call (strike approximately 530). Each contract corresponds to 100 shares of SPY. At initiation, the position generates a net premium credit of approximately \$627,000, reflecting the elevated implied volatility of downside puts relative to upside calls. This structure expresses a bullish directional bias combined with a short-volatility and skew-selling viewpoint.

The economic foundation of the trade lies in its asymmetric option exposures. Because the trader is short the lower-strike put, the position carries short downside convexity, negative gamma, and negative vega at low strikes. Meanwhile, the long upside call provides long gamma and long vega at higher strikes. Since the implied volatility of the 95% put is significantly greater than that of the 105% call, the overall structure is short implied volatility. The trade also earns positive theta, as the time decay of the short put exceeds that of the long call. Consequently, the trade performs well when SPY remains stable or appreciates, when downside skew compresses, or when implied volatility remains unchanged or declines. Conversely, the trade performs poorly during sharp downside moves, volatility spikes, or periods in which the left tail of the return distribution realizes strongly.

Using a Black–Scholes–Merton framework, we compute the mark-to-market P&L one week prior to expiration across a range of spot scenarios. The results are highly asymmetric: a +15% increase in SPY produces approximately +\$5.66 million, an unchanged spot level yields +\$471,000 (primarily from theta decay), while a –15% decline results in –\$4.48 million. These outcomes reflect the substantial upside convexity and the significant downside tail risk inherent in the structure.

To assess whether the market environment on 9 April 2025 supports entering this trade, we perform a historical analysis using daily SPY data from January 2023 to April 2025, a period comprising 554 observations. We compute 21-day realized volatility, 1-month at-the-money implied volatility, the volatility risk premium ($VRP = IV - RV$), and the 95%–105% risk-reversal skew. Summary statistics for the sample are reported in Table 1.

Statistic	Realized Vol	ATM IV (1m)	VRP	Skew (RR)
Mean	13.35%	14.07%	0.73%	6.68%
Std. Dev	5.50%	3.95%	3.93%	1.49%
Min	5.76%	9.33%	-24.9%	2.53%
Median	12.31%	12.95%	0.93%	6.56%
Max	50.69%	42.32%	11.1%	15.85%

Table 1: Historical Volatility and Skew Characteristics (Jan 2023–Apr 2025)

The volatility risk premium on 9 April 2025 lies in the **0.5th percentile** of its historical distribution. This indicates that implied volatility is extremely cheap relative to realized volatility. Such highly negative VRP readings typically occur during periods when realized volatility spikes sharply (for example, due to macroeconomic shocks), whereas implied volatility reacts more gradually. A VRP in the bottom percentile suggests that selling volatility is poorly compensated in absolute terms and is therefore unattractive.

In contrast, the risk-reversal skew on the trade date lies in the **99.8th percentile** of the sample. This indicates that downside puts are extraordinarily expensive relative to upside calls—one of the most extreme skew environments observed in the dataset. In such settings, selling downside insurance and purchasing upside convexity is strongly favored from a relative-value standpoint.

Overall, the signals deliver a mixed message. From a relative-value perspective, the skew is exceptionally rich and strongly supports the trade. From an absolute-volatility perspective, the VRP is extremely low and argues against shorting volatility. Therefore, entering the trade requires conviction in a stable or bullish near-term outlook and a tolerance for downside tail exposure.

Finally, the maximum loss and gain characteristics of the trade further illustrate its asymmetry. The expiry payoff is given by

$$\Pi(S_T) = -1000 \times 100 \cdot \max(K_{\text{put}} - S_T, 0) + 1000 \times 100 \cdot \max(S_T - K_{\text{call}}, 0).$$

If SPY rallies substantially above the call strike, the position behaves like a leveraged long call. For example, a +15% rally produces roughly \$5.66 million in profit, and the upside is theoretically unbounded. On the downside, if SPY were to fall to zero, the payoff would be

$$\Pi_{\max \text{ loss}} \approx -1000 \times 100 \cdot 480 + 627,000 = -\$47.4 \text{ million.}$$

Even moderate declines generate large losses—for instance, a 15% drop yields approximately −\$4.48 million one week before expiry. Thus, the structure resembles a heavily leveraged short put complemented by an upside kicker: upside potential is meaningful though roughly linear in spot beyond the call strike, while downside risk is large, convex, and potentially catastrophic. These properties underscore the importance of entering the position only when skew compensation is extreme or when a strong directional view is held.